Catch Me If You Can: Quantifying Outfield Shifting Impact for Baseball Fans

Team 150

Abstract

Defense is the black sheep of baseball, overlooked and underappreciated. There is a lot that goes into making a catch in the outfield, and most fans only recognize the last few seconds of a play as to whether it was a catch or an error. However, there are many cases where a spectacular play was made, but it looked like a routine catch. A lot of plays are properly defended before the batter even makes contact with the pitch. This project calculates catch probability on balls hit to the outfield compared to the catch probability of the same ball where the outfielder would start from at the average position at contact. Using the SMT-provided player and ball positioning data, calculate the optimal route distance, hang time, attack angle (the angle at which the fielder needs to turn to field the ball), and the descent angle of the ball to create a catch probability system that can be applied to both actual results of the play, and a what-if scenario where the player had not shifted in preparation for the batter. This allows for a catch-probability based presentation that can help the fans understand the impact of outfield defensive shifting.

Intro

With the player and ball positioning data provided by SMT being the unique adventure provided by this competition, I wanted to do something that capitalized on this opportunity. I decided to look into how to expand off of it and turn it into something that would benefit a scoreboard. Catch Probability has been the cornerstone of the outfield portion of MLB's Statcast metric OAA (outs above average) for evaluating defense since 2017. Originally built just to evaluate hang time, optimal route distance, and quickly edited a few months later to include

attack angle (direction), it provides an easy-to-understand summary of what an outfielder had to do to make a play. Unfortunately, the most important parts of making a catch are usually over before the audience has even had time to look at the appropriate fielder. The starting position, the reaction to the ball, by the time fans are watching all that is usually left is for the fielder to run in a straight line. That is why a visual presentation of the direct impact outfield shifting has would be an asset for any ball club (see fig. 1).



Fig 1. Example Graphic. Outfielder silhouette repurposed from Vecteezy

Data and Analysis

Using Python and expanding on the very basics of hang time and optimal route distance in one of the tutorials, research from Tangotiger's blog included a list of parameters he wanted to try in Statcast's catch probability model but also showed the effectiveness of adding in the decent angle of the ball. After exploring this and seeing an improvement in the logistic regression model I had built, a deeper dive into Tango's thoughts on catch probability suggested that the angle of direction from the fielder to the ball would be another good place to go. Filtering for hits over 175ft to get balls only in the outfield, and ignoring mishandled baseballs or balls that hit walls to better isolate our targeted opportunities. Then discerning the average starting location of the

outfielders, split by the handedness of the batter to create an average starting location, set to isolate the impact of shifting from the difference in the optimal route distance and attack angle variables. We see that 8.1% of plays in the outfield carry a 10% or greater change in catch probability from the model based on deviation from the average starting location and 1.7% of plays to the outfield see a 50% or more change in probability from shifts. (see fig. 2).

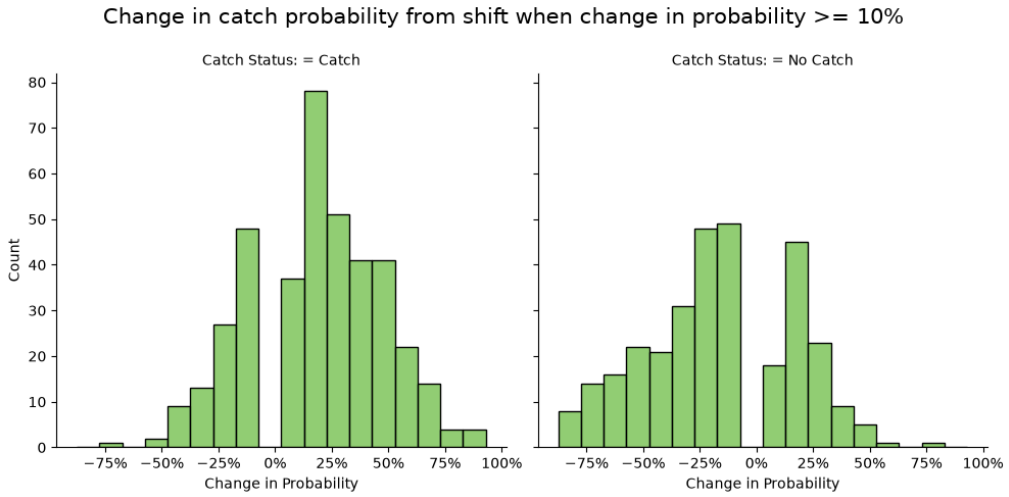


Fig 2. Change in catch probability from shift split by catch status where change in probability >= 10% based on actual vs average starting location.

Discussion

Implementation in game would be to highlight when a catch was made successfully via shifting that would have been hard to notice with the naked eye. Teams could use this as an opportunity to highlight the unseen work of the contributors behind the scenes. An example graphic using a real play in the data provided by SMT (see fig. 1). Gives the audience context as to why a shift is important, and some of the easy-to-understand underlying metrics behind a stat like catch probability.

Next Steps

In the future, there is a lot that I would like to expand on. First, the scoreboard mock-up could have comparable routes and animations of both the actual and shifted route being run. This would be a great way to establish clearly to the audience what is being gained by shifting. I would also love to both improve the catch probability model in terms of code efficiency, but also to add in some more variables to increase its capabilities. Lastly, I think building out RE24 and establishing a run value provided by the change in catch probability would be an excellent way to emphasize the value catching a few extra outfield balls has on the game of baseball.

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